pop quiz - torques - equilibrium

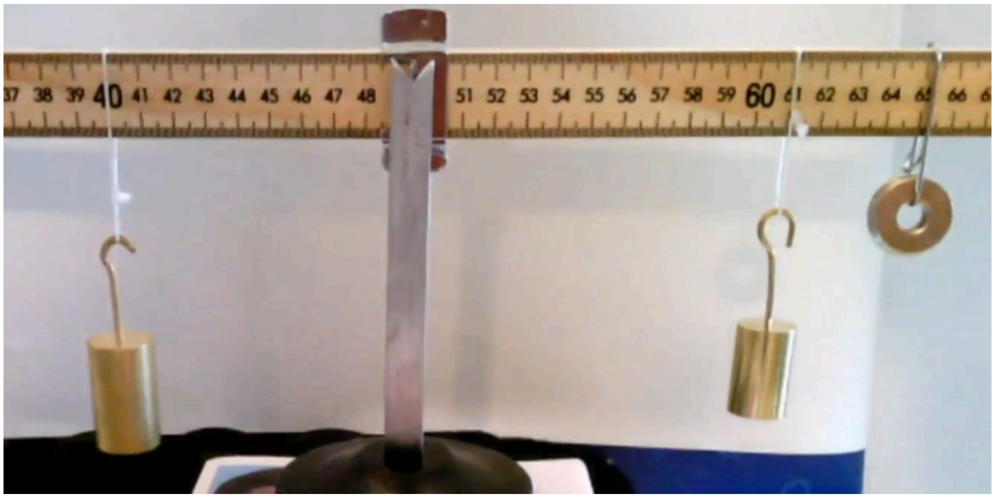
This is a preview of the published version of the quiz

Started: Apr 29 at 7:45pm

Quiz Instructions

Talk to each other

Question 1 14.3 pts



Here is a static equilibrium. from left to right. you see that m1 is 10cm from the fulcrum - left side

m2 is at 11cm from the fulcrum right side. and m3 is is at .... cm from the fulcrum right side. count the cm.

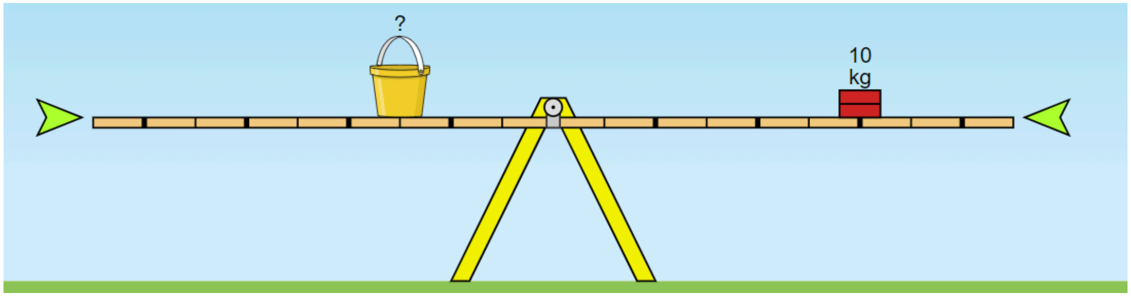
m1 is 50g. m2 is 25 g. so what is m3 ?

Don't ask me. Use sum(CW) = sum(CCW)

Don't convert g to N. don't convert cm to m

- 15g
- 25g
- 50g
- 5g

Question 2 14.3 pts

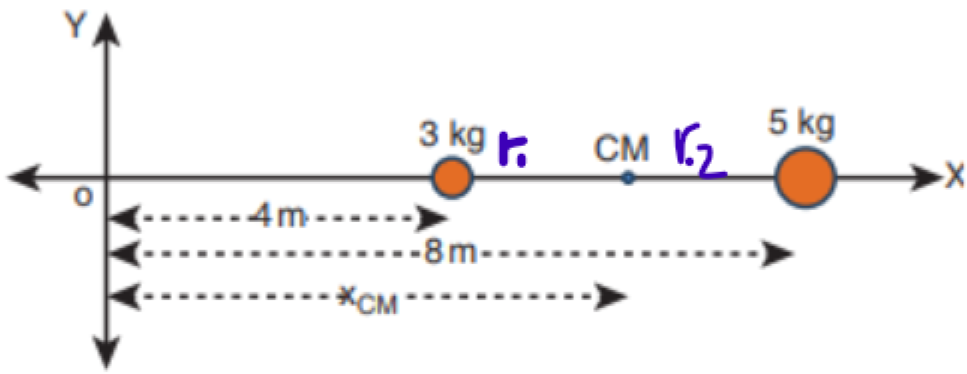


what is the mass of the bucket?

- 20kg

- ○ 10kg
- ○ 30kg
- ○ 5kg

Question 3 14.3 pts



Find the position of the center of mass. x1=4m and m1 = 3kg. x2=8m and m2 = 5kg.

If you place a fulcrum at CM it will be in equilibrium (3r1 = 5r2). With a little bit of algebra you get the equation for the center of mass:

Xcm = (x1 m1 + x2 m2) / (m1+m2)

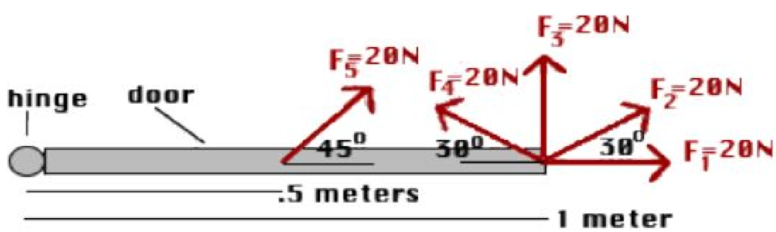
- ○ 6.5cm
- ○ 6cm
- ○ 7cm
- ○ 5cm

Question 4 14.3 pts

Its an equilibrium because

- ◯ his center of mass is above the line connecting the hands
- ◯ he is held up by an insisible string. can't be otherwise
- ◯ he is extremely muscular
- ◯ this is a photo generated by Grok (X AI). can't be

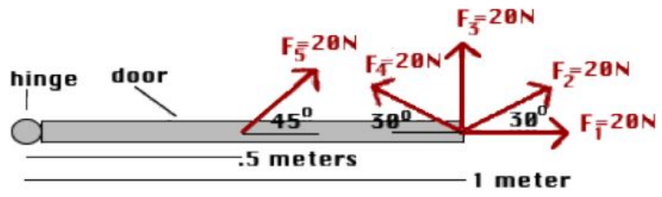
Question 5 14.3 pts



The torque of the force F1 is

- ◯ 0
- ◯ 20N.m
- ◯ 10 N.m
- ◯ I miss the class

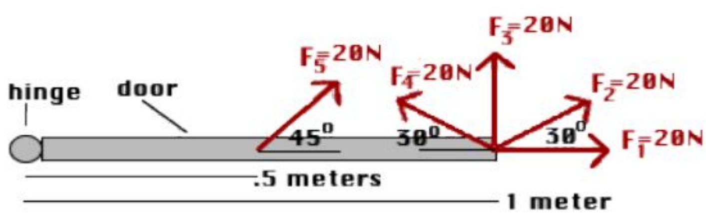
Question 6 14.3 pts



The torque of the force F3 is

- 20N.m
- 0
- 10 N.m
- 5N.m

Question 7 14.3 pts



The torque of F5 is

hint: Use the perpendicular component (ry-component of F5).

- 7N.m
- 14 N.m
- 20N.m
- 0

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